Iraq National Single Window (NSW) Technical Integration Specifications

Interfacing Manual for Maritime Ports, Border Terminals, ASYCUDA & CBI

INTERFACE ARCHITECTURE REGISTER:

â¢ Document Reference: IDG-NSW-INT-09
â¢ Target Corridors: Umm Qasr Port (North/South), Trebil, Ibrahim Khalil, Safwan
â¢ System Counterparts: ASYCUDA World (Customs), BUNA, CBI RTGS, Port Operating Systems
â¢ Protocol Version: REST / JSON / XML EDIFACT v4.2 & gRPC Streaming

1. End-to-End Corridor Flow Overview

The Iraq National Single Window streamlines international trade through five digital steps:
Step 1: Manifest Pre-Arrival Declaration (Shipping Line / Carrier to NSW Gateway).
Step 2: Automated Single Electronic Customs Declaration (ASYCUDA World Interchange).
Step 3: Algorithmic Multi-Agency Risk Assessment (Green, Yellow, Red Channeling).
Step 4: Instant Financial Clearance via Central Bank (CBI) Sovereign Gateway.
Step 5: Digital Release Order with Holographic QR Gate Verification.

2. Pre-Arrival Cargo Manifest Schema (Endpoint: /api/v1/nsw/manifest)

POST /api/v1/nsw/manifest - Headers: Authorization: Bearer <mTLS_JWT>

Payload Structure:

```
 {
   "manifestId": "MNF-2026-BSR-008912",
   "carrier": "National Iraqi Maritime Transport Co",
   "portOfEntry": "IQ-UQR (Umm Qasr South)",
   "eta": "2026-11-04T08:00:00Z",
   "containersCount": 420,
   "cargoItems": [
    {"hsCode": "8471.30", "declaredValueUSD": 1420000, "weightKg": 18450}
   ]
 }
```

3. Central Bank RTGS Settlement Endpoint (Endpoint: /api/v1/nsw/settle)

Upon risk validation, tariffs and fees are deducted in Iraqi Dinars (IQD):

â¢ Automatic generation of Sadad institutional billing tokens.

â¢ Immediate credit acknowledgment to the General Directorate of Customs sub-treasury.

â¢ Cryptographic proof-of-payment token appended to digital container release order.

4. Automated Risk Engine & AI Manifest Auditing

â¢ Green Channel (70-80%): Instant clearance under 2 hours without physical inspection.
â¢ Yellow Channel (15-20%): Automated document verification and invoice price audit.
â¢ Red Channel (5-10%): Mandatory non-intrusive container scanner and physical inspection.

5. Terminal Gate-Out & Holographic QR Verification

The terminal exit gate utilizes encrypted, time-limited digital release tokens:
â¢ Handheld border guard scanners verify token authenticity offline against sovereign root cert.
â¢ Instant container gate-out status reported to Ministry of Transport real-time telemetry.
â¢ Overall port dwell time reduced from 18 days to under 48 hours nationwide.

INTERFACING CERTIFICATION:

[APPROVED] Directorate General of Customs - Republic of Iraq

[CERTIFIED] Port & Border Corridors Taskforce - Iraq Digital Gateway (IDG)